

Business Analytics Report

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1. Dataset Overview

Total Rows: 125751

Total Columns: 16

Missing Values: 0

2. Column Information

- Transaction Date: object
- Sales Type: object
- Is Return: int64
- Supplier: object
- Product No: object
- Product Description: object
- Product Division: object
- Product Category: object
- Product Subcategory: object
- Product Segment: object
- Store: object
- Sales Channel: object
- Qty Sold: float64
- Sales Amount: float64
- Cogs: float64

3. SQL Analysis Results

Query 1 - Top Categories:

- Rank 1: Full Price
- Rank 2: Promo
- Rank 3: Markdown Tier 1
- Rank 4: Markdown Tier 2
- Rank 5: Clearance

4. Statistical Metrics

- Total Rows: 125751
- Total Columns: 16
- Missing Values: 0

- Duplicate Rows: 0
- Is Return - Total: 7491
- Is Return - Average: 0.06
- Is Return - Median: 0.0
- Is Return - Std Dev: 0.24
- Qty Sold - Total: 116995.0
- Qty Sold - Average: 0.93
- Qty Sold - Median: 1.0
- Qty Sold - Std Dev: 0.59
- Sales Amount - Total: 10482110.25
- Sales Amount - Average: 83.36
- Sales Amount - Median: 90.0
- Sales Amount - Std Dev: 83.66
- Cogs - Total: 5847751.1
- Cogs - Average: 46.5
- Cogs - Median: 50.05
- Cogs - Std Dev: 43.6

5. AI Analysis & Recommendations

1. Business Domain:

This dataset appears to be related to the retail industry, focusing on sales and returns data for various products. It likely comes from an e-commerce platform, a brick-and-mortar store, or a combination of both. The presence of a "Sales Channel" column and the detailed product information suggest that the dataset might be from a company that operates through multiple sales channels.

2. Key Findings:

- * The average return rate is very low at 0.06%, indicating that the company has a relatively low incidence of returns. This might be due to the quality of products or effective sales practices.
- * The average sales amount per transaction is \$83.36, which suggests that transactions are relatively small and frequent.
- * The standard deviation of sales amount is high at \$83.66, implying that there is a significant amount of variation in sales values.
- * The number of transactions per transaction day is very high at 1.04. However, this might not be directly comparable to other businesses, and more context is needed to understand its significance.

3. Business Risks:

- * Inconsistent Sales Patterns: The high standard deviation of sales amount and the varying sales patterns across different products and times might indicate inconsistent sales patterns. This could lead to difficulties in forecasting sales and inventory management.
- * Returns and Inventory Management: Although the return rate is low, there might still be returns that could lead to overstocking or inventory discrepancies. This could affect cash flow and sales growth.
- * Channel Performance: The presence of multiple sales channels might lead to variations in sales performance across

channels. Monitoring and optimizing performance across channels will be crucial to maximize revenue.

4. Recommendations:

1. **Analyze Sales Channel Performance:** Segment the data by sales channels and analyze sales performance for each channel. This will help the company understand which channels are generating the most revenue and optimize marketing efforts accordingly.
2. **Improve Inventory Management:** Implement a data-driven inventory management system that incorporates sales forecasts and real-time inventory levels. This will help minimize overstocking and inventory discrepancies.
3. **Optimize Sales and Pricing Strategies:** Use data analysis to identify areas where prices can be optimized to improve sales. This might involve analyzing competitor pricing, sales trends, and product profitability.
4. **Monitor Customer Behavior:** Incorporate customer behavior data (e.g., purchase frequency, retention rates, product preferences) to gain a better understanding of customer needs and preferences. This will enable targeted marketing strategies and improve customer satisfaction.

6. Forecast

- May 2026: 11,116.16
- Jun 2026: 10,982.19
- Jul 2026: 10,848.22
- Aug 2026: 10,714.25
- Sep 2026: 10,580.27
- Oct 2026: 10,446.30
- Nov 2026: 10,312.33
- Dec 2026: 10,178.35
- Jan 2027: 10,044.38
- Feb 2027: 9,910.41
- Mar 2027: 9,776.44
- Apr 2027: 9,642.46